

What is the best geometry for minimization of MMD?

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August 21, 2026

1 Introduction

We focus on MMD due to its interpretation as the worse-case integration error in the unit ball of a RKHS.

Notations: For a probability measure π , assume that $\int k(x, x) d\pi(x) < \infty$. We define the associated kernel integral operator $\mathcal{T}_\pi : L_2(\pi) \rightarrow L_2(\pi)$ by $(\mathcal{T}_\pi f)(x) = \int k(x, y) f(y) d\pi(y)$ for π -almost every x , where, as usual, elements of $L_2(\pi)$ are understood up to π -almost-everywhere equivalence.

2 Gradient Flows

Gradient flows provide a principled framework for evolving a curve of probability measures $(\mu_t)_{t \geq 0}$ along the direction of steepest descent of an objective functional $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \rightarrow \mathbb{R}$. To define what constitutes the “steepest” direction, however, one must first equip $\mathcal{P}(\mathbb{R}^d)$ with a metric $d(\mu, \nu)$ —different choices of d induce different geometries on $\mathcal{P}(\mathbb{R}^d)$ and therefore lead to different gradient-flow dynamics, even when the objective $\mathcal{F}$ is fixed.

In this paper, we focus on the minimization of the (squared) MMD objective, i.e., $\mathcal{F}_{\text{MMD}} = \frac{1}{2} \text{MMD}^2(\cdot, \pi)$, due to its wide application in numerical integration and generative modelling [Li et al., 2017]. We summarize several resulting gradient flows from common choices of geometry d and discuss the convergence behaviour of these dynamics, i.e., whether $\lim_{t \rightarrow \infty} \mathcal{F}(\mu_t) \rightarrow 0$. In this section, we focus on the gradient flows at the population level, and defer their practical realizations through finite-particle approximations to the next section.

Before we proceed to introduce various gradient flows, we first need to introduce the first variation. The first variation of $\mathcal{F}_{\text{MMD}}$ is $\mathcal{F}'_{\text{MMD}} : x \mapsto \int k(x, y) d(\mu - \pi)(y)$.

Wasserstein One of the most commonly used geometries is the Wasserstein-2 geometry, defined on $\mathcal{P}_2(\mathbb{R}^d)$, the set of probability measures with finite second moment. In this case, the metric d is chosen to be the Wasserstein-2 distance, $d(\mu, \nu) = W_2(\mu, \nu) := \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int \|x - y\|^2 d\gamma(x, y) \right)^{\frac{1}{2}}$. Intuitively, $W_2(\mu, \nu)$ represents the minimum quadratic cost required to move the mass of μ to match ν . Under this geometry, the Wasserstein gradient flow (W-GF) of $\mathcal{F}_{\text{MMD}}$ is

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla \mathcal{F}'_{\text{MMD}}[\mu_t]). \quad (\text{W-GF})$$

Fisher-Rao Another classical geometry on the space of probability measures is the Fisher–Rao geometry. With a slight abuse of notation, let $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ denote probability measures that admit densities, also denoted by μ and ν , with respect to Lebesgue measure. The Fisher–Rao distance is given by $d(\mu, \nu) = d_{\text{FR}}(\mu, \nu) := 2 \arccos \left(\int \sqrt{\mu(x)\nu(x)} dx \right)$. The resulting Fisher–Rao gradient flow (FR-GF) of $\mathcal{F}_{\text{MMD}}$ is:

$$\partial_t \mu_t = -\mu_t \left(\mathcal{F}'_{\text{MMD}}[\mu_t] - \int \mathcal{F}'_{\text{MMD}}[\mu_t] d\mu_t \right). \quad (\text{FR-GF})$$

Wasserstein Fisher-Rao (W-GF) evolves a distribution by continuously transporting mass through space, whereas (FR-GF) redistributes the density locally without requiring mass to move along spatial paths. An interpolation between these two dynamics is the Wasserstein–Fisher–Rao gradient flow (WFR-GF), allowing the probability measure to adapt through both particle movement and mass redistribution. For $\alpha, \beta > 0$,

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'_{\text{MMD}}[\mu_t]) - \beta \mu_t \left(\mathcal{F}'_{\text{MMD}}[\mu_t] - \int \mathcal{F}'_{\text{MMD}}[\mu_t] d\mu_t \right). \quad (\text{WFR-GF})$$

Interaction Force Transport More recently, an alternative geometry based on the MMD has been studied in [Zhu \[2024\]](#), [Zhu and Mielke \[2024\]](#), [Gladin et al. \[2024\]](#). For a kernel k and probability measure μ , recall the associated kernel integral operator $\mathcal{T}_\mu : L^2(\mu) \rightarrow L^2(\mu)$. Also recall that $\mathcal{F}'_{\text{MMD}} = \int k(\cdot, y) d(\mu - \pi)(y) = \mathcal{T}_\mu(1 - d\mu/d\pi)$. The resulting Interaction-Force Transport (IFT) gradient flow of $\mathcal{F}$ is given by [\[Gladin et al., 2024\]](#):

$$\partial_t \mu_t = -\mu_t \mathcal{T}_{\mu_t}^{-1} \mathcal{F}'_{\text{MMD}}[\mu_t] = -(\mu_t - \pi). \quad (\text{IFT-GF})$$

The above dynamics seems only well-defined when the inverse operator $\mathcal{T}_{\mu_t}^{-1}$ cancel out with the operator $\mathcal{T}_{\mu_t}$ in $\mathcal{F}'_{\text{MMD}}$. Since we only focus on $\mathcal{F}_{\text{MMD}}$ in this paper, we leave an extension to more general $\mathcal{F}$ to future work.

Wasserstein Interaction Force Transport Similarly as (WFR-GF), an interpolation of WGF and IFT-GF can also be constructed

$$\partial_t \mu_t = \alpha \nabla \cdot (\mu_t \nabla \mathcal{F}'[\mu_t]) - \beta \mathcal{T}^{-1} \mathcal{F}'[\mu_t]. \quad (\text{WIFT-GF})$$

It has been proved that [\[Gladin et al., 2024\]](#), for both IFT-GF and WIFT-GF, $\text{MMD}^2(\mu_t, \pi) \leq e^{-2\beta t} \cdot \text{MMD}^2(\mu_0, \pi)$.

3 Particle Descent

If $\mathcal{F}$ is the MMD distance $\mathcal{F}(\cdot) = \text{MMD}^2(\cdot, \pi)$, then the first variation $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot) d(\mu - \pi) = \mathcal{T}(\mu - \pi)$. Then, we can derive the particle descent scheme of the corresponding gradient flows.

Let $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)} \delta_{x_t^{(i)}}$ be a weighted empirical measure at iteration t . Write $\mathbf{X}_t = (x_t^{(1)}, \dots, x_t^{(N)})$ and $\mathbf{w}_t = (w_t^{(1)}, \dots, w_t^{(N)})$. Denote $m_\pi = \int k(x, \cdot) d\pi$ as the kernel mean embedding. Denote the kernel-mean residual and its vectorized counterpart as,

$$h_{\mathbf{X}, \mathbf{w}}(x) := \sum_{j=1}^N w^{(j)} k(x^{(j)}, x) - m_\pi(x) \in \mathbb{R}, \quad h_{\mathbf{X}, \mathbf{w}}(\mathbf{v}) = [h_{\mathbf{X}, \mathbf{w}}(v^{(1)}), \dots, h_{\mathbf{X}, \mathbf{w}}(v^{(N)})]^\top \in \mathbb{R}^N.$$

Define

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}, \pi \right).$$

Euclidean geometry The most straightforward approach to optimizing the particle locations and weights is to apply gradient descent jointly to $\mathbf{X}$ and $\mathbf{w}$. For consistency with the update schemes introduced later, we adopt a Gauss–Seidel-type block gradient descent scheme, in which the particle locations are updated first and the weights are subsequently updated using the new locations $\mathbf{X}_{t+1}$. In particular, let $\tau > 0$ be the step size. We consider $\mathbf{X}_{t+1} = \mathbf{X}_t - \tau \nabla_{\mathbf{X}} E(\mathbf{X}_t, \mathbf{w}_t)$ and $\mathbf{w}_{t+1} = \mathbf{w}_t - \tau \nabla_{\mathbf{w}} E(\mathbf{X}_t, \mathbf{w}_t)$. Expanding the gradients yields the following explicit update scheme: for each $i \in [N]$,

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau w_t^{(i)} \nabla h_{\mathbf{X}_t, \mathbf{w}_t} \left(x_t^{(i)} \right), \quad w_{t+1}^{(i)} = w_t^{(i)} - \tau h_{\mathbf{X}_{t+1}, \mathbf{w}_t} \left(x_{t+1}^{(i)} \right) \quad (1)$$

Wasserstein Fisher-Rao geometry

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau \alpha \nabla h_{\mathbf{X}_t, \mathbf{w}_t} \left(x_t^{(i)} \right), \quad w_{t+1}^{(i)} = w_t^{(i)} - \tau \beta w_t^{(i)} h_{\mathbf{X}_{t+1}, \mathbf{w}_t} \left(x_{t+1}^{(i)} \right). \quad (2)$$

Wasserstein IFT geometry The locations are first updated as the same as above, however, the weight update scheme require knowledge of the true density π . In reality, the weights are updated instead via an implicit minimizing scheme.

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau \alpha \nabla h_{\mathbf{X}_t, \mathbf{w}_t} \left(x_t^{(i)} \right), \quad (3)$$

$$\mathbf{w}_{t+1} = \arg \min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x_{t+1}^{(i)}}, \pi \right) + \frac{1}{2(\beta\tau)} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x_{t+1}^{(i)}}, \mu_{N,t} \right) \right\}. \quad (4)$$

Interestingly, letting $K_{t+1} = K(\mathbf{X}_{t+1}, \mathbf{X}_{t+1}) \in \mathbb{R}^{N \times N}$ be the Gram matrix, the weight problem is

$$\min_{\mathbf{w} \in \Delta^N} \left\{ \frac{1}{2} \mathbf{w}^\top K_{t+1} \mathbf{w} - m_\pi(\mathbf{X}_{t+1})^\top \mathbf{w} + \frac{1}{2\eta} (\mathbf{w} - \mathbf{w}_t)^\top K_{t+1} (\mathbf{w} - \mathbf{w}_t) \right\},$$

By definition of kernel quadrature, there is $m_\pi(\mathbf{X}_{t+1}) = K_{t+1} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$. And the solution is, when ignoring the simplex constraint, $\mathbf{w}_{t+1} = \frac{1}{1+\eta} \mathbf{w}_t + \frac{\eta}{1+\eta} \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$ which can be equivalently written as,

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{1}{1 + (\tau\beta)} K_{t+1}^{-1} (h_{\mathbf{X}_{t+1}, \mathbf{w}_t}(\mathbf{X}_{t+1})).$$

To state the particle updates compactly, Table 1 gives the location and weight updates explicitly for the three geometries, direct Euclidean gradient descent, and MSIP [Belhadji et al., 2026]. In the WFR scheme, the transport step precedes the weight step, which is evaluated at the updated particle locations. The WFR weight update is obtained by applying forward Euler discretization to the Fisher–Rao reaction term in Eq. (WFR-GF).

The Wasserstein row uses uniform initial weights, which remain equal to $1/N$ under transport; arbitrary initial weights would likewise remain fixed. The centering term in the WFR update preserves $\sum_{i=1}^N w_t^{(i)} = 1$, while nonnegativity additionally requires $\tau\beta$ to be sufficiently small. The MMD row is the WIFT particle implementation used here: it combines the Wasserstein location

Table 1: Location and weight updates for MMD minimization under Wasserstein, WFR, and MMD geometries, together with direct Euclidean gradient descent and MSIP.

Geometry or method	Location and weight updates
Wasserstein	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\mathbf{X}_t, \mathbf{w}_t}(x_t^{(i)}),$ $w_{t+1}^{(i)} = w_t^{(i)} = \frac{1}{N}.$
WFR	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\mathbf{X}_t, \mathbf{w}_t}(x_t^{(i)}),$ $w_{t+1}^{(i)} = w_t^{(i)} - \tau\beta w_t^{(i)} \left[h_{\mathbf{X}_{t+1}, \mathbf{w}_t}(x_{t+1}^{(i)}) - \sum_{j=1}^N w_t^{(j)} h_{\mathbf{X}_{t+1}, \mathbf{w}_t}(x_{t+1}^{(j)}) \right].$
MMD (IFT)	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\mathbf{X}_t, \mathbf{w}_t}(x_t^{(i)}),$ $\mathbf{w}_{t+1} = \frac{1}{1+\eta}\mathbf{w}_t + \frac{\eta}{1+\eta}\mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$
Direct GD	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha w_t^{(i)} \nabla h_{\mathbf{X}_t, \mathbf{w}_t}(x_t^{(i)}),$ $w_{t+1}^{(i)} = w_t^{(i)} - \tau\beta h_{\mathbf{X}_t, \mathbf{w}_t}(x_t^{(i)}).$
MSIP	For a squared-exponential kernel, $x_{t+1}^{(i)} = (1-\tau)x_t^{(i)} + \frac{\tau}{w_{\text{KQ}}^{(i)}(\mathbf{X}_t)} \sum_{j=1}^N [K(\mathbf{X}_t)^{-1}]_{ij} \int x k(x, x_t^{(j)}) d\pi(x),$ $\mathbf{w}_{t+1} = \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}).$

update with the IFT weight update in Eqs. (??)–(??). Its displayed closed form is obtained when the simplex constraint in Eq. (??) is ignored. The direct-gradient row performs simultaneous Euclidean descent in both $\mathbf{X}$ and $\mathbf{w}$; its location update differs from the Wasserstein update by the factor $w_t^{(i)}$. This weight update is unconstrained and must be projected if the iterates are required to remain in Δ^N . In contrast, MSIP is a damped fixed-point method for the stationary conditions of the unbalanced WFR flow, rather than a time discretization of the normalized WFR flow in Eq. (WFR-GF). Its update requires $K(\mathbf{X}_t)$ to be invertible and the kernel quadrature weights to be positive.

Definition 3.1 (Weighted Euclidean Geometry). *Let $m \in \mathbb{R}$ and let $G \in \mathbb{R}^{m \times m}$ be a symmetric positive definite matrix. We equip the space $\mathbb{R}^m$ with the G -weighted Euclidean inner product $\langle u, v \rangle_G := u^\top G v$ and with the induced norm $\|u\|_G^2 := u^\top G u$. For a differentiable function $f : \mathbb{R}^m \rightarrow \mathbb{R}$, its gradient with respect to the G -weighted Euclidean geometry, denoted by $\text{grad}_G f$, is defined as: for any $u \in \mathbb{R}^m$,*

$$\left. \frac{d}{dt} f(x + tu) \right|_{t=0} = \langle \text{grad}_G f(x), u \rangle_G.$$

Equivalently, $\text{grad}_G f(x) = G^{-1} \nabla f(x)$ where ∇f denotes the standard Euclidean gradient.

Proposition 3.2 (Smoothness under Euclidean and weighted Euclidean geometries). *Let $\mathbf{w} \in \Delta^N$ with $w^{(i)} > 0$ for every i and $\sum_{i=1}^N w^{(i)} = 1$. Suppose that k is symmetric and twice continuously differentiable, with $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1 < \infty$ and $\sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} = C_2 < \infty$.*

Define

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}, \pi \right).$$

Then $E(\cdot, \mathbf{w})$ is $\max_{1 \leq i \leq N} w^{(i)}(2C_1 + C_2)$ -smooth under the standard Euclidean geometry; and is $(2C_1 + C_2)$ -smooth under the $\mathbf{w}$ -weighted Euclidean geometry. Moreover, $E(\mathbf{X}, \cdot)$ is $\lambda_{\max}(K(\mathbf{X}, \mathbf{X}))$ -smooth under the standard Euclidean geometry; and is 1-smooth under the $K(\mathbf{X}, \mathbf{X})$ -weighted Euclidean geometry.

From the above proposition, we see that the right smoothness notion to consider is the weighted Euclidean geometry instead of the standard Euclidean geometry.

The direct KQ update instantaneously minimizes the MMD over the weights at each particle configuration, whereas the IFT update is a proximal tracking scheme that interpolates between the previous weights and the instantaneous KQ solution. The latter sacrifices instantaneous optimality in exchange for a controlled and stable evolution of the weights, consistent with the underlying continuous-time IFT dynamics.

4 Convergence

Theorem 4.1 (Monotone improvement). *At iteration $t \in \mathbb{N}$, let $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)} \delta_{x_t^{(i)}}$ with $\mathbf{X}_t = (x_t^{(1)}, \dots, x_t^{(N)})$, and $\mathbf{w}_t = (w_t^{(1)}, \dots, w_t^{(N)})$.*

Assume that k is twice continuously differentiable, that $K(\mathbf{X}_t)$ is positive definite for every t , and that the weights remain uniformly positive, namely $w_t^{(i)} \geq \underline{w} > 0$ for every i, t . Assume moreover that, for every admissible $\mathbf{w}$, the map $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w})$ has an L_X -Lipschitz gradient on a set containing all iterates. If $\tau\alpha \leq 2\underline{w}/L_X$, then for every $t \geq 0$,

$$\text{MMD}(\mu_{N,t+1}, \pi) \leq \text{MMD}(\mu_{N,t}, \pi). \quad (5)$$

Lemma 4.2 (Uniform control of the kernel quadrature residual). *Suppose that the assumptions of Theorem 4.1 hold and that the initial weights are chosen as $\mathbf{w}_0 = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$. Then, for every $t \geq 0$,*

$$\text{MMD} \left(\sum_{i=1}^N w_{\text{KQ}}^{(i)}(\mathbf{X}_t) \delta_{x_t^{(i)}}, \pi \right) \leq \text{MMD} \left(\sum_{i=1}^N w_{\text{KQ}}^{(i)}(\mathbf{X}_0) \delta_{x_0^{(i)}}, \pi \right).$$

Theorem 4.3 (Convergence to the kernel quadrature error). *Suppose that the assumptions of Theorem 4.1 hold, and that the initial weights are chosen as $\mathbf{w}_0 = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$. Then, the WIFT iterates satisfy*

$$\text{MMD}^2(\mu_{N,t}, \pi) \leq \frac{1}{(1+\eta)^{2t}} \text{MMD}^2(\mu_{N,0}, \pi) + \left(1 - \frac{1}{(1+\eta)^{2t}} \right) \text{MMD}^2 \left(\sum_{i=1}^N w_{\text{KQ}}^{(i)}(\mathbf{X}_0) \delta_{x_0^{(i)}}, \pi \right).$$

Remark 4.4. *Theorem 4.3 separates the optimization error from the statistical error of the initial kernel quadrature rule. In particular, if $\text{MMD}(\sum_{i=1}^N w_{\text{KQ}}^{(i)}(\mathbf{X}_0) \delta_{x_0^{(i)}}, \pi) = O_{\mathbb{P}}(N^{-s/d})$, then*

$$\text{MMD}(\mu_{N,t}, \pi) \leq \frac{1}{(1+\eta)^t} \text{MMD}(\mu_{N,0}, \pi) + O_{\mathbb{P}}(N^{-s/d}).$$

Thus, the optimization error decays geometrically until it reaches the kernel quadrature error floor. If, in addition, $\text{MMD}(\mu_{N,0}, \pi) = O_{\mathbb{P}}(1)$, then choosing $t \geq \frac{s}{d \log(1+\eta)} \log N$ is sufficient to obtain $\text{MMD}(\mu_{N,t}, \pi) = O_{\mathbb{P}}(N^{-s/d})$.

5 Proofs

Proof of Proposition 3.2. Recall that

$$E(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \sum_{i,j=1}^N w^{(i)} w^{(j)} k(x^{(i)}, x^{(j)}) - \sum_{i=1}^N w^{(i)} m_\pi(x^{(i)}) + \text{const.}$$

For any $\mathbf{u} = (u^{(1)}, \dots, u^{(N)}) \in (\mathbb{R}^d)^N$, direct differentiation gives

$$\begin{aligned} \nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}] &= \sum_{i,j=1}^N w^{(i)} w^{(j)} (u^{(i)})^\top \nabla_{11}^2 k(x^{(i)}, x^{(j)}) u^{(i)} \\ &\quad + \sum_{i,j=1}^N w^{(i)} w^{(j)} (u^{(i)})^\top \nabla_{12}^2 k(x^{(i)}, x^{(j)}) u^{(j)} - \sum_{i=1}^N w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)}. \end{aligned}$$

Since $m_\pi(x) = \int k(x, y) d\pi(y)$ and π is a probability measure, we have

$$\|\nabla^2 m_\pi(x)\|_{\text{op}} = \left\| \int \nabla_{11}^2 k(x, y) d\pi(y) \right\|_{\text{op}} \leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{11}^2 k(x, y)\|_{\text{op}} = C_1.$$

Using $\sum_{j=1}^N w^{(j)} = 1$, it follows that

$$\left| \sum_{i,j=1}^N w^{(i)} w^{(j)} (u^{(i)})^\top \nabla_{11}^2 k(x^{(i)}, x^{(j)}) u^{(i)} - \sum_{i=1}^N w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)} \right| \leq 2C_1 \sum_{i=1}^N w^{(i)} \|u^{(i)}\|^2.$$

For the mixed term,

$$\begin{aligned} \left| \sum_{i,j=1}^N w^{(i)} w^{(j)} (u^{(i)})^\top \nabla_{12}^2 k(x^{(i)}, x^{(j)}) u^{(j)} \right| &\leq \sup_{x, y \in \mathbb{R}^d} \|\nabla_{12}^2 k(x, y)\|_{\text{op}} \left(\sum_{i=1}^N w^{(i)} \|u^{(i)}\| \right)^2 \\ &= C_2 \left(\sum_{i=1}^N w^{(i)} \|u^{(i)}\| \right)^2. \end{aligned}$$

By Cauchy-Schwarz and $\sum_{i=1}^N w^{(i)} = 1$, we have $\left(\sum_{i=1}^N w^{(i)} \|u^{(i)}\| \right)^2 \leq \sum_{i=1}^N w^{(i)} \|u^{(i)}\|^2$. Therefore,

$$|\nabla_{\mathbf{X}}^2 E(\mathbf{X}, \mathbf{w})[\mathbf{u}, \mathbf{u}]| \leq (2C_1 + C_2) \sum_{i=1}^N w^{(i)} \|u^{(i)}\|^2 = (2C_1 + C_2) \|\mathbf{u}\|_{\mathbf{w}}^2.$$

The last step holds by the definition of $\mathbf{w}$ -weighted Euclidean geometry in Definition 3.1. This proves smoothness under the weighted Euclidean geometry.

For the standard Euclidean geometry, since $\sum_{i=1}^N w^{(i)} \|u^{(i)}\|^2 \leq (\max_{1 \leq i \leq N} w^{(i)}) \sum_{i=1}^N \|u^{(i)}\|^2 = (\max_{1 \leq i \leq N} w^{(i)}) \|\mathbf{u}\|_2^2$. The proof is concluded. $\square$

Proof of Theorem 4.1. For fixed $\mathbf{X}$ and $\mathbf{w}$, let $g^{(i)}(\mathbf{X}, \mathbf{w}) = \sum_{j=1}^N w^{(j)} \nabla_2 k(x^{(j)}, x^{(i)}) - \nabla m_\pi(x^{(i)})$. Define the MMD energy by

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}, \pi \right), \quad (6)$$

Differentiating the MMD energy with respect to $x^{(i)}$ and using symmetry of the kernel gives $\nabla_{x^{(i)}} \mathcal{E}(\mathbf{X}, \mathbf{w}) = w^{(i)} g^{(i)}(\mathbf{X}, \mathbf{w})$. Hence, the position update can be written as $x_{t+1}^{(i)} = x_t^{(i)} - \frac{\tau\alpha}{w_t^{(i)}} \nabla_{x^{(i)}} \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t)$. Thus, the Wasserstein substep is a diagonally preconditioned gradient step for $\mathbf{X} \mapsto \mathcal{E}(\mathbf{X}, \mathbf{w}_t)$. By the L_X -smoothness of $\mathcal{E}(\cdot, \mathbf{w}_t)$, the descent lemma gives

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_t) &\leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) + \sum_{i=1}^N \left\langle \nabla_{x^{(i)}} \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t), x_{t+1}^{(i)} - x_t^{(i)} \right\rangle + \frac{L_X}{2} \sum_{i=1}^N \|x_{t+1}^{(i)} - x_t^{(i)}\|^2 \\ &= \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) - \tau\alpha \sum_{i=1}^N \left(w_t^{(i)} - \frac{L_X \tau\alpha}{2} \right) \|g^{(i)}(\mathbf{X}_t, \mathbf{w}_t)\|^2. \end{aligned} \quad (7)$$

Since $w_t^{(i)} \geq \underline{w}$ and $\tau\alpha \leq 2\underline{w}/L_X$, we have $w_t^{(i)} - L_X \tau\alpha/2 \geq 0$. Therefore, $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_t) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t)$. This proves that the particle update in Eq. (??) is decreasing the MMD value.

We next consider the weight update. For fixed $\mathbf{X}$, the dependence of the MMD energy on $\mathbf{w}$ is quadratic: $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \mathbf{w}^\top K(\mathbf{X}) \mathbf{w} - m_\pi(\mathbf{X})^\top \mathbf{w} + \text{const}$. Since $K(\mathbf{X})$ is positive definite, the unique minimizer is $\mathbf{w}_{\text{KQ}}(\mathbf{X}) = K(\mathbf{X})^{-1} m_\pi(\mathbf{X})$. Completing the square gives

$$\mathcal{E}(\mathbf{X}, \mathbf{w}) = \mathcal{E}(\mathbf{X}, \mathbf{w}_{\text{KQ}}(\mathbf{X})) + \frac{1}{2} (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X}))^\top K(\mathbf{X}) (\mathbf{w} - \mathbf{w}_{\text{KQ}}(\mathbf{X})). \quad (8)$$

By the weight update,

$$\mathbf{w}_{t+1} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}) = \frac{1}{1+\eta} (\mathbf{w}_t - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \quad (9)$$

On the other hand, for fixed particle locations $\mathbf{X}_{t+1}$, the KQ decomposition gives

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{2} (\mathbf{w}_{t+1} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))^\top K_{t+1} (\mathbf{w}_{t+1} - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \end{aligned} \quad (10)$$

Substituting (9) into (10) yields

$$\begin{aligned} \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{2(1+\eta)^2} (\mathbf{w}_t - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))^\top K_{t+1} (\mathbf{w}_t - \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \\ = \frac{1}{(1+\eta)^2} [\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_t) - \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1}))]. \end{aligned} \quad (11)$$

Since $(1+\eta)^{-2} \leq 1$ and $\mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})$ minimizes $\mathbf{w} \mapsto \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w})$, it follows that $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_t)$. Combining the two substeps yields

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) \leq \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_t) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t). \quad (12)$$

Hence, $\mathcal{E}(\mathbf{X}_t, \mathbf{w}_t)$ is nonincreasing, and the proof is concluded. $\square$

Proof of Lemma 4.2. Recall the definition that $\mathcal{E}(\mathbf{X}, \mathbf{w}) = \frac{1}{2} \text{MMD}^2 \left(\sum_{i=1}^N w^{(i)} \delta_{x^{(i)}}, \pi \right)$. For any fixed particle locations $\mathbf{X}_t$, the kernel quadrature weights $\mathbf{w}_{\text{KQ}}(\mathbf{X}_t)$ minimize the MMD energy over the weights. Hence, $\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t)$. On the other hand, by the monotonicity of the WIFT iteration established in Theorem 4.1, we have $\mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}_0)$. Since $\mathbf{w}_0 = \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)$, we have $\mathcal{E}(\mathbf{X}_0, \mathbf{w}_0) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$. Combining the previous inequalities gives

$$\mathcal{E}(\mathbf{X}_t, \mathbf{w}_{\text{KQ}}(\mathbf{X}_t)) \leq \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}_0) = \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (13)$$

The proof is thus concluded. $\square$

Proof of Theorem 4.3. From the exact contraction of the weight update in Eq. (11), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})). \quad (14)$$

By Lemma 4.2, $\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{\text{KQ}}(\mathbf{X}_{t+1})) \leq \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0))$, and therefore

$$\mathcal{E}(\mathbf{X}_{t+1}, \mathbf{w}_{t+1}) \leq \frac{1}{(1+\eta)^2} \mathcal{E}(\mathbf{X}_t, \mathbf{w}_t) + \left(1 - \frac{1}{(1+\eta)^2}\right) \mathcal{E}(\mathbf{X}_0, \mathbf{w}_{\text{KQ}}(\mathbf{X}_0)). \quad (15)$$

The proof is concluded. $\square$

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